



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Brett Perlas
3-30-2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- The goal of this project is to become familiar with Python Data Science techniques by applying them to a real world scenario
- We hope to figure out certain key metrics such as success rate of landings and determine correlation between successful landing and things like payload, launch site/origin, etc

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Collected historical Falcon 9 launch data via both REST API calls to the official SpaceX API and web scraping applicable tables from Wikipedia using BeautifulSoup.
- Perform data wrangling
 - Cleaned the dataset by handling missing values, calculating the payload mass mean, and converting categorical variables (like Orbits, Launch Sites, and landing outcomes) into numerical/binary features using One-Hot Encoding.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Data sets were collected by utilizing both REST API calls and web scraping
- First, we used Python to obtain a Pandas dataframe populated by the output of API calls to the official SpaceX Rest API to the appropriate and relevant end points
- Additionally, we used Python module BeautifulSoup to web scrape relevant data from Wikipedia's page containing historical SpaceX launch data and incorporated it into our dataframe.

Data Collection – SpaceX API

- Utilized API endpoints to call on and receive JSON response data that we normalized to import into a pandas dataframe
- GitHub URL:
https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/94d3aca53e6201ae0e37496808af7ec59ecbe527/Final_Report/jupyter-labs-spacex-data-collection-api.ipynb

Process Flow:

Request Object -> API endpoint -> JSON
Response Object -> Pandas DataFrame

Data Collection - Scraping

- Utilized BeautifulSoup to extract the targeted data from the correct tables on Wikipedia into a pandas dataframe
- GitHub URL:
https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/94d3aca53e6201ae0e37496808af7ec59ecbe527/Final_Report/jupyter-labs-webscraping.ipynb

Process Flow:

Wikipedia HTML Tables -> BeautifulSoup Object -> Parsing/Processing -> Pandas DataFrame -> CSV Export (optional)

Data Wrangling

- Cleaned the dataset by handling missing values, calculating the payload mass mean, and converting categorical variables (like Orbits, Launch Sites, and landing outcomes) into numerical/binary features using One-Hot Encoding.
- Removed NaNs -> Replace missing numeric values with appropriate mean values -> Create categorical variables as needed
- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- Utilized SQL queries and Pandas to explore the data.
- Created visual scatterplots and bar charts using Matplotlib and Seaborn to identify relationships between payload mass, flight number, orbit type, and landing success.
- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/edadataviz.ipynb

EDA with SQL

- Summary of SQL Queries:

- `SELECT DISTINCT Launch_Site from SPACEXTBL`
- `SELECT Launch_Site FROM SPACEXTBL WHERE (Launch_Site LIKE 'CCA%') LIMIT 5;`
- `SELECT SUM(PAYLOAD_MASS__KG_) as payload_mass_total FROM SPACEXTBL`
- `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE (Booster_Version = 'F9 v1.1')`
- `SELECT MIN(DATE) FROM SPACEXTBL WHERE (Landing_Outcome = 'Success (ground pad)')`
- **`SELECT DISTINCT`** `Booster_Version` **`FROM`** `SPACEXTBL`
- **`WHERE`** `(Landing_Outcome = 'Success (drone ship)')` **`AND`** `((PAYLOAD_MASS__KG_ > 4000) AND (PAYLOAD_MASS__KG_ < 6000))`
- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/jupyter-labs-eda-sql-coursera_sqlite.ipynb

EDA with SQL (continued)

- Summary of SQL Queries:

- **SELECT DISTINCT** Booster_Version **FROM** SPACEXTBL **WHERE** PAYLOAD_MASS__KG_ = (**SELECT** **MAX**(PAYLOAD_MASS__KG_) **FROM** SPACEXTBL)
- **SELECT** Mission_Outcome, **COUNT**(Mission_Outcome) **FROM** SPACEXTBL **GROUP BY** Mission_Outcome
- **SELECT** substr(Date, 6,2) **as** Month, Landing_Outcome, Booster_Version, Launch_Site, substr(Date,0,5) **as** Year **FROM** SPACEXTBL **WHERE** ((Landing_Outcome = 'Failure (drone ship)') **AND** (Year = '2015'))
- **SELECT** Landing_Outcome, **COUNT**(Landing_Outcome) **as** Count **FROM** SPACEXTBL **WHERE** (Date > '2010-06-04') **AND** (Date < '2017-03-20') **GROUP BY** Landing_Outcome **ORDER BY** Count **DESC**

- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Added Folium Map Objects
 - Added 4 marker objects to represent the SpaceX Launch Site locations.
 - Clusters were used for visual flow so that geographically close together launches can appear as fewer points on a zoomed out map
 - Coastline coordinates were also mapped out as marker objects to aid in determining distances from launch sites to coastline.
 - Polyline objects were used to display distance to coastline from launch site visually, in addition to plotting distance between sites and closest city, railway, or highway.
- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/94d3aca53e6201ae0e37496808af7ec59ecbe527/Final_Report/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

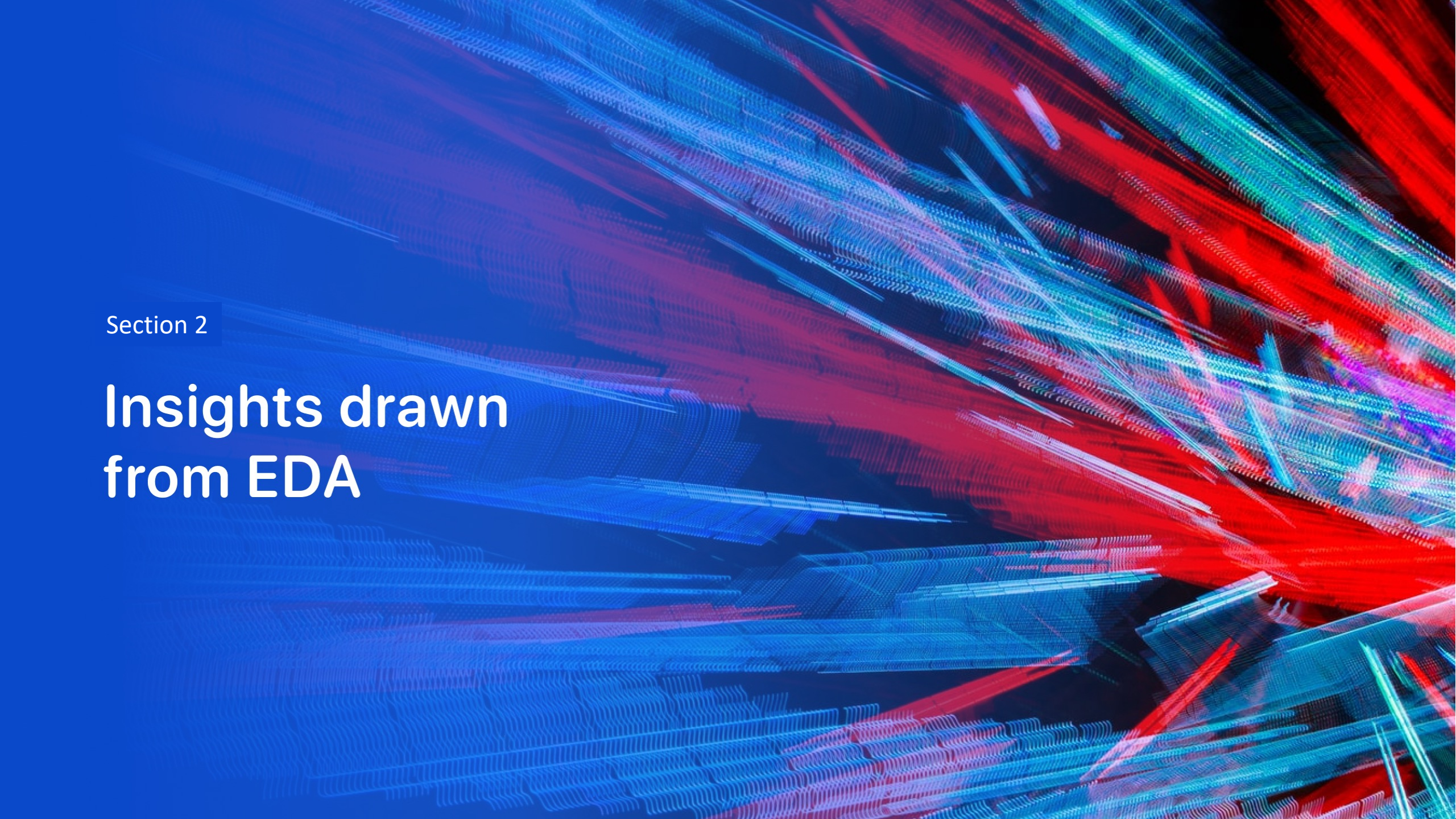
- Primary Interaction Widget is a payload slider
 - This allows filtering launch datapoints by payload weight
 - Visualizations include a Pie Chart and Scatterplot to show success rate by launch site for the given upper and lower bounds of payload weight
- GitHub URL: https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/94d3aca53e6201ae0e37496808af7ec59ecbe527/Final_Report/spacex-dash-app.py

Predictive Analysis (Classification)

- Standardized the data and split it into training (80%) and testing (20%) sets.
- Built four machine learning classification models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Used GridSearchCV with 10-fold cross-validation to tune hyperparameters and find the optimal version of each model
- Add the GitHub [URL:https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb](https://github.com/DocMidnite/coursera-Data-Science-Projects/blob/83558790424e241c874dcd8638157a07971d347e/Final_Report/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb)

Results

- Exploratory data analysis results
 - Success Rate improves over time / with experience (practice makes perfect)
 - Orbit Type has a significant impact on mission success
 - Larger payloads have a higher success rate
- Interactive analytics demo in screenshots (see slides below in next section for dashboard screenshots)
- Predictive analysis results
 - **Model Accuracies:** Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) all tied for the highest test-set accuracy at **83.33%**.
 - **Best Model: KNN** was selected as the optimal model due to having the most consistent Cross-Validation score during training.
 - **Confusion Matrix:** The top models successfully predicted 15 out of 18 test cases, flawlessly identifying 100% of the successful landings, with only 3 false positives.

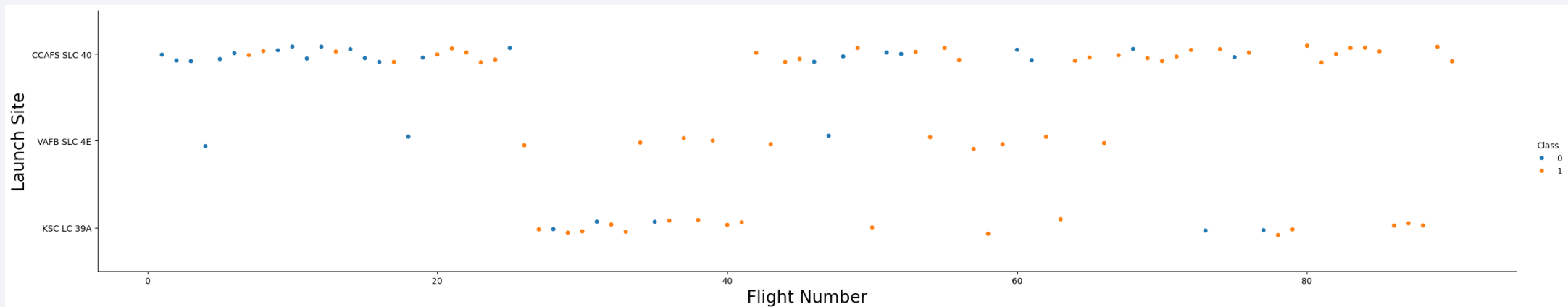
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

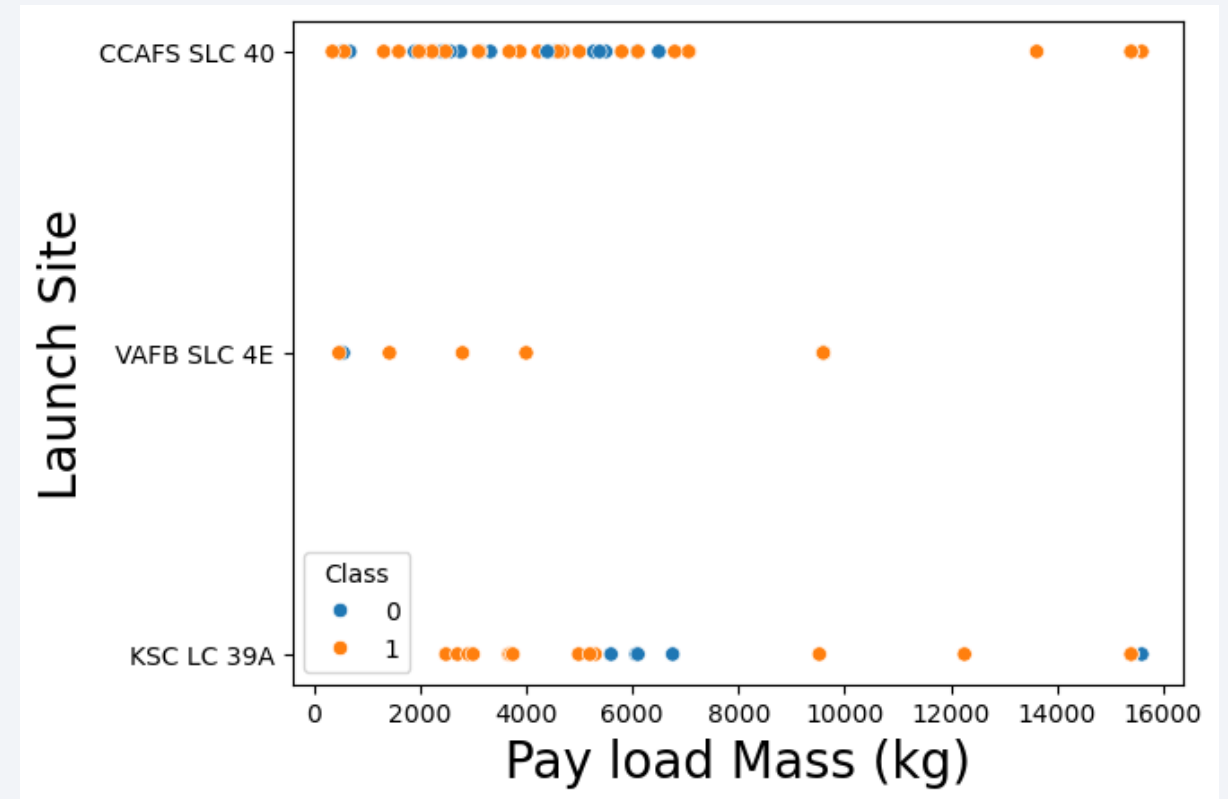
Flight Number vs. Launch Site

- Flight numbers increase over time while failures decrease over time
- Higher success and higher flight numbers are correlated
- Earlier utilized launch sites had higher failure rates while launch sites established later had higher success rates
- Practice Makes Perfect



Payload vs. Launch Site

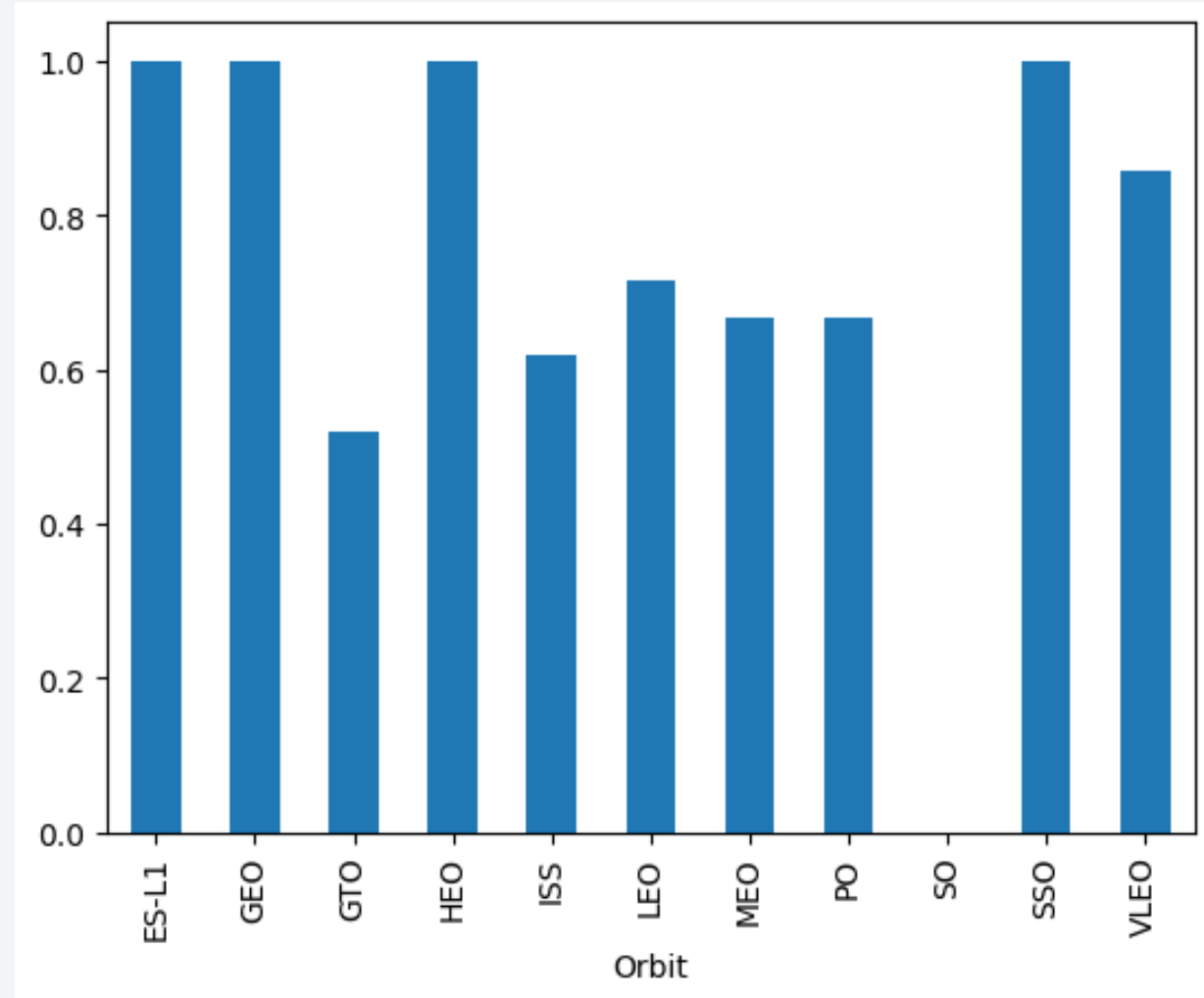
- Main Site for Heavy Payloads: CCAFS SLC 40
- Main sites for lighter loads: *VAFB and KSC*
- Heavier payloads have higher success rates than lower payload launches
- Lighter payloads also tend to be associated with earlier flights which were less successful (see previous slide)



Success Rate vs. Orbit Type

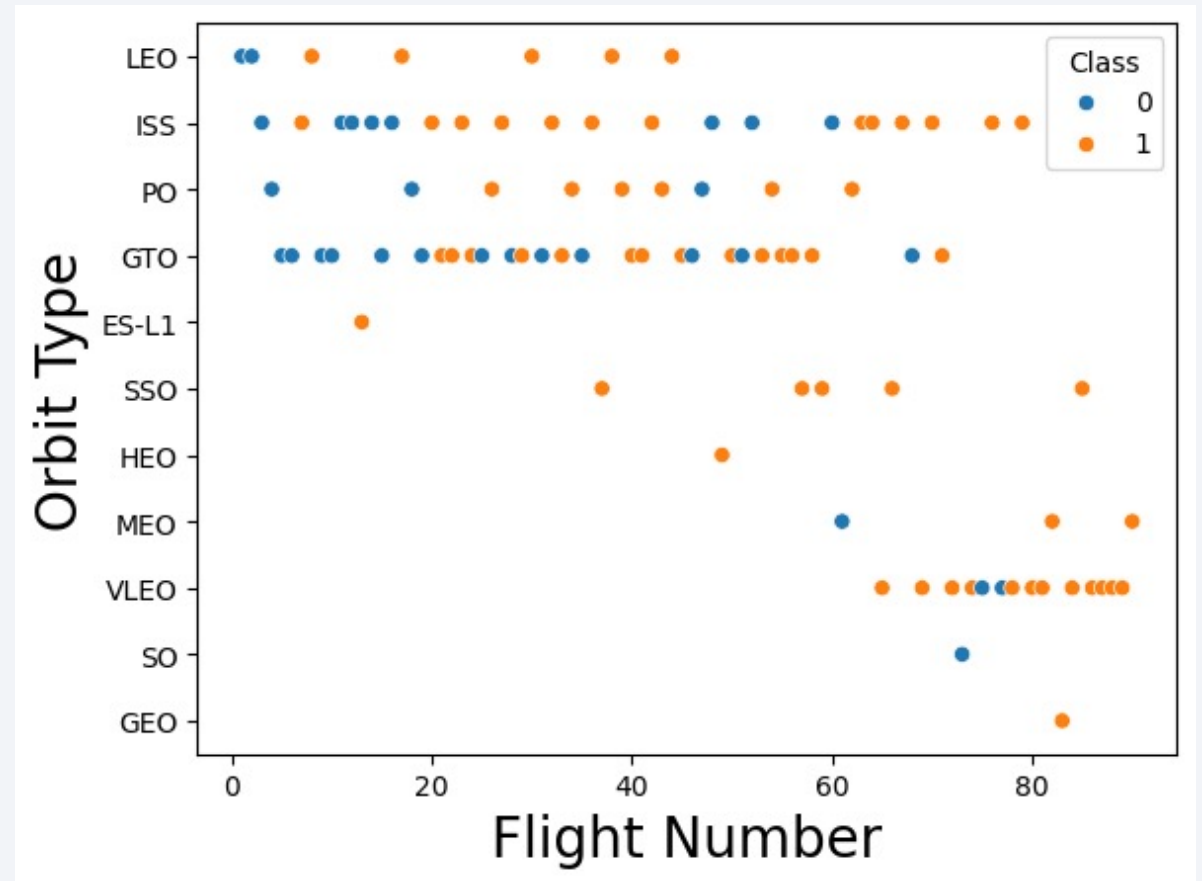
- Success Rates Are Highest for 4 Types of Orbits:

- ES-L1
- GEO
- HEO
- SSO



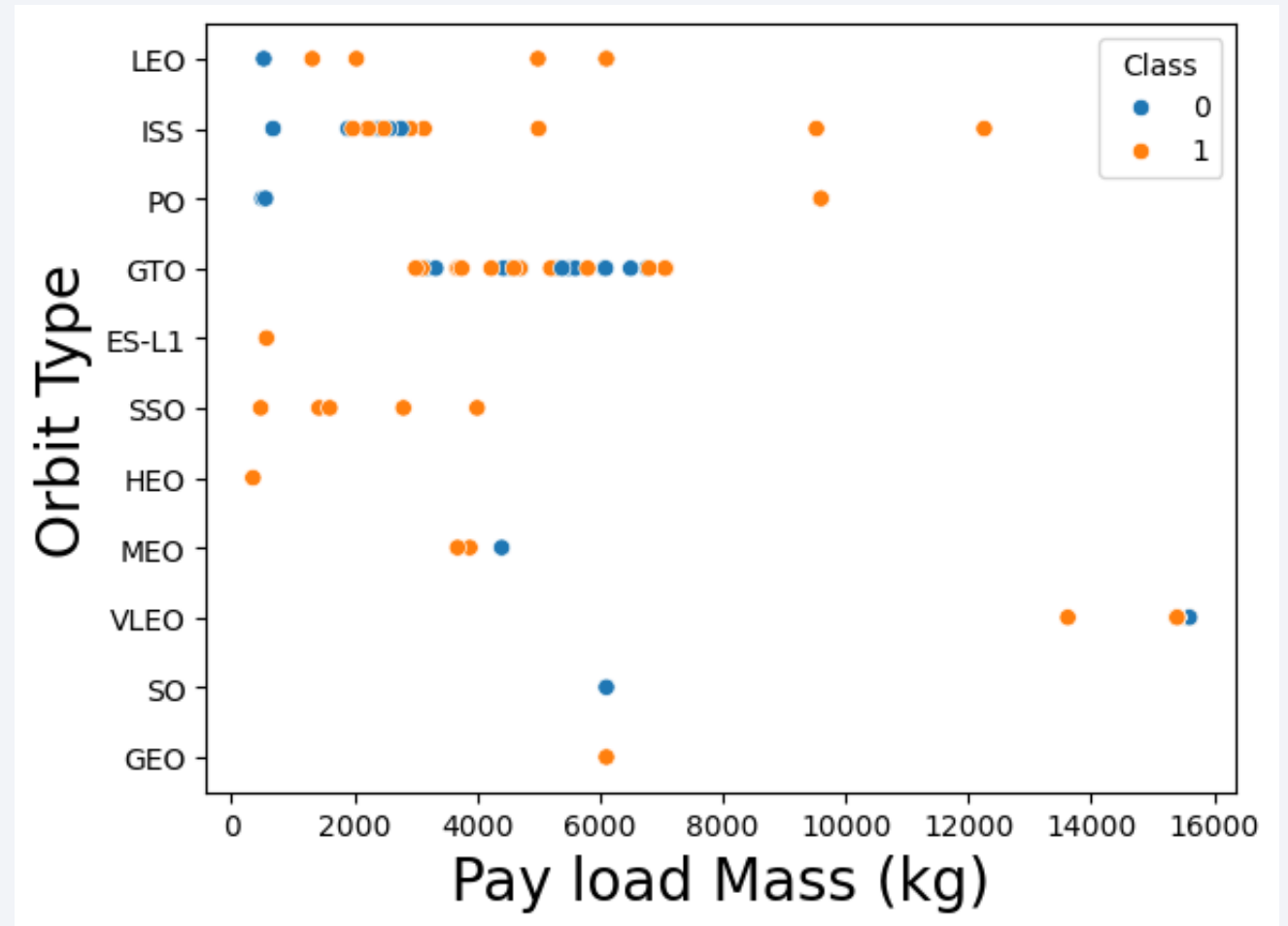
Flight Number vs. Orbit Type

- Trend of higher success rate over time persists when disaggregated by orbit type
- VLEO appears to be the current favored orbit type with both more frequency and higher success rates over time
- LEO may have been phased out entirely
- ESL-1, HEO, and Geo are rare so their 1.0 success rates are based on 1 just data point each / highly dubious. Likewise for SO's 0.0 success rate



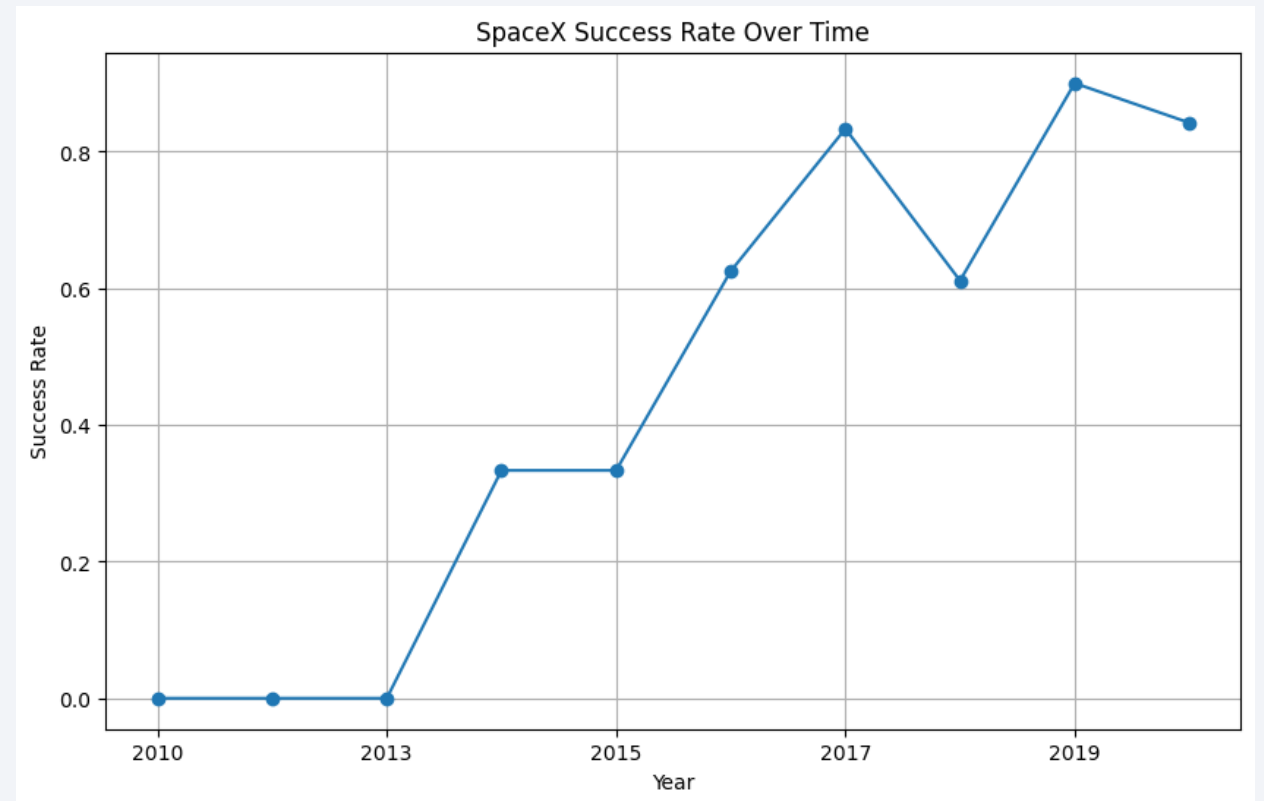
Payload vs. Orbit Type

- Heaviest payloads tend to utilize VLEO orbit
- ISS tends to be utilized for lighter payloads
- GTO tends to be utilized for mid range payloads 3k-8k
- Some 100% success rate orbit types have very few data points regardless of payload mass (see last slide)



Launch Success Yearly Trend

- Average success rate has steadily increased over time in a clear trend
- This reinforces the practice makes perfect idea touched upon when we factored in flight numbers (which increase chronologically over time)



All Launch Site Names

Query: SELECT DISTINCT Launch_Site from SPACEXTBL

- SpaceX Launch Sites:

CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Query: `SELECT Launch_Site FROM SPACEXTBL WHERE (Launch_Site LIKE 'CCA%') LIMIT 5;`
- Note that we did not specify “distinct” so there will be duplicates:

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Query: `SELECT SUM(PAYLOAD__MASS__KG_) as payload_mass_total FROM SPACEXTBL`
- Result:

payload_mass_total
619967

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

- Query:

- `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE (Booster_Version = 'F9 v1.1')`

- Result:

<code>VG(PAYLOAD_MASS__KG_)</code>

2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Query:

- `SELECT MIN(DATE) FROM SPACEXTBL WHERE (Landing_Outcome = 'Success (ground pad)')`

- Result:

MIN(DATE)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Query:
- **SELECT DISTINCT** Booster_Version **FROM** SPACEXTBL
 - **WHERE** (Landing_Outcome = 'Success (drone ship)') **AND** ((PAYLOAD_MASS__KG_ > 4000) **AND** (PAYLOAD_MASS__KG_ < 6000))

- Results:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Query:
 - **SELECT** Mission_Outcome, **COUNT**(Mission_Outcome) **FROM** SPACEXTBL **GROUP BY** Mission_Outcome

- Results:

Mission_Outcome	COUNT(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Query:
 - **SELECT DISTINCT** Booster_Version **FROM** SPACEXTBL **WHERE** PAYLOAD_MASS__KG_ = (**SELECT MAX**(PAYLOAD_MASS__KG_) **FROM** SPACEXTBL)
 - We utilize a subquery to use the max payload values from the data itself as a parameter in the master query
 - Results (on right):

Booster_Version

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Query:
 - **SELECT** substr(Date, 6,2) **as Month**, Landing_Outcome, Booster_Version, Launch_Site, substr(Date,0,5) **as Year**
 - **FROM** SPACEXTBL **WHERE** ((Landing_Outcome = 'Failure (drone ship)') **AND** (Year = '2015'))
- Results:

Month	Landing_Outcome	Booster_Version	Launch_Site	Year
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Query:
 - SELECT** Landing_Outcome, **COUNT**(Landing_Outcome) **as Count** **FROM** SPACEXTBL **WHERE** (Date > '2010-06-04') **AND** (Date < '2017-03-20') **GROUP BY** Landing_Outcome **ORDER BY** Count **DESC**
- Results (on right):

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue area on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

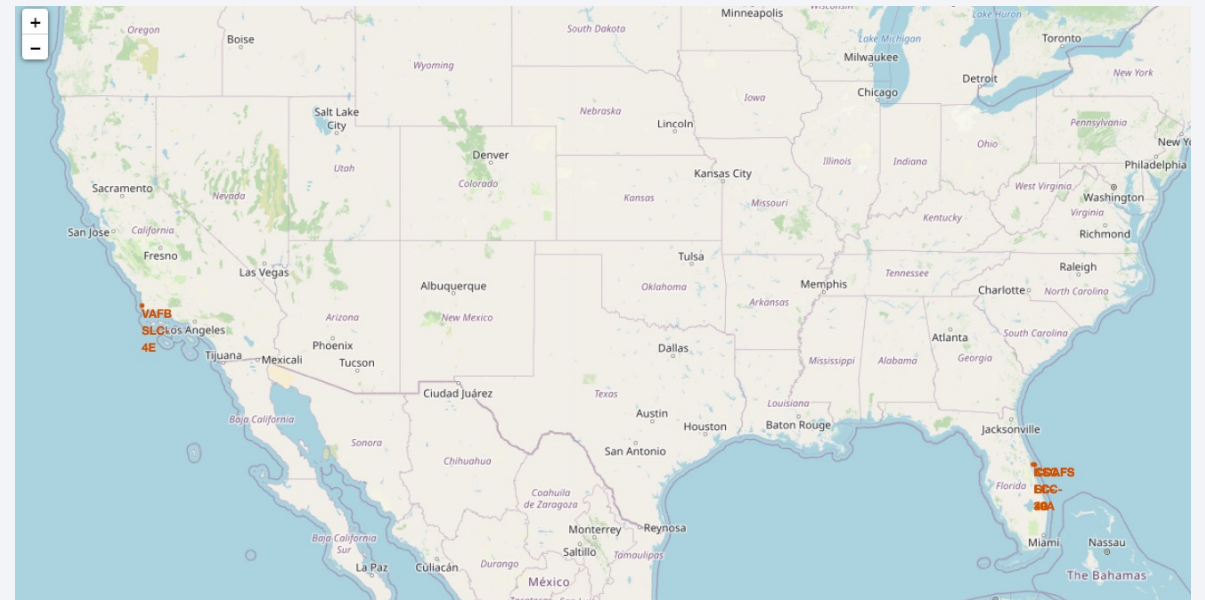
Section 3

Launch Sites Proximities Analysis

Map of All SpaceX Launch Sites

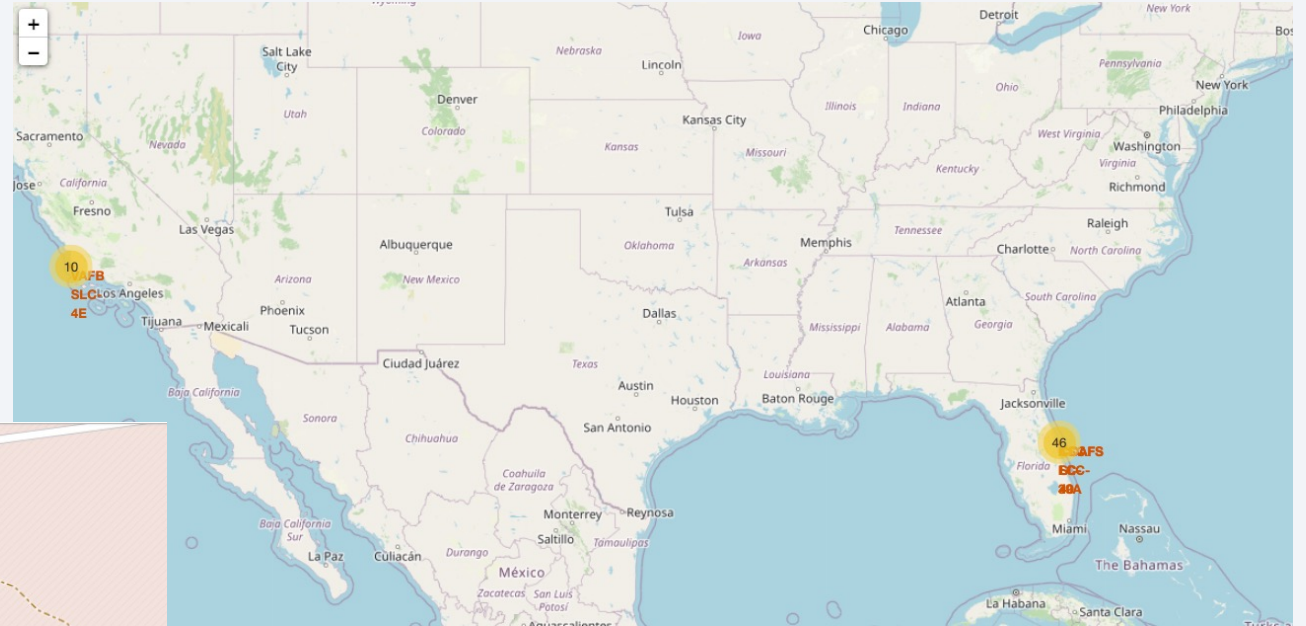
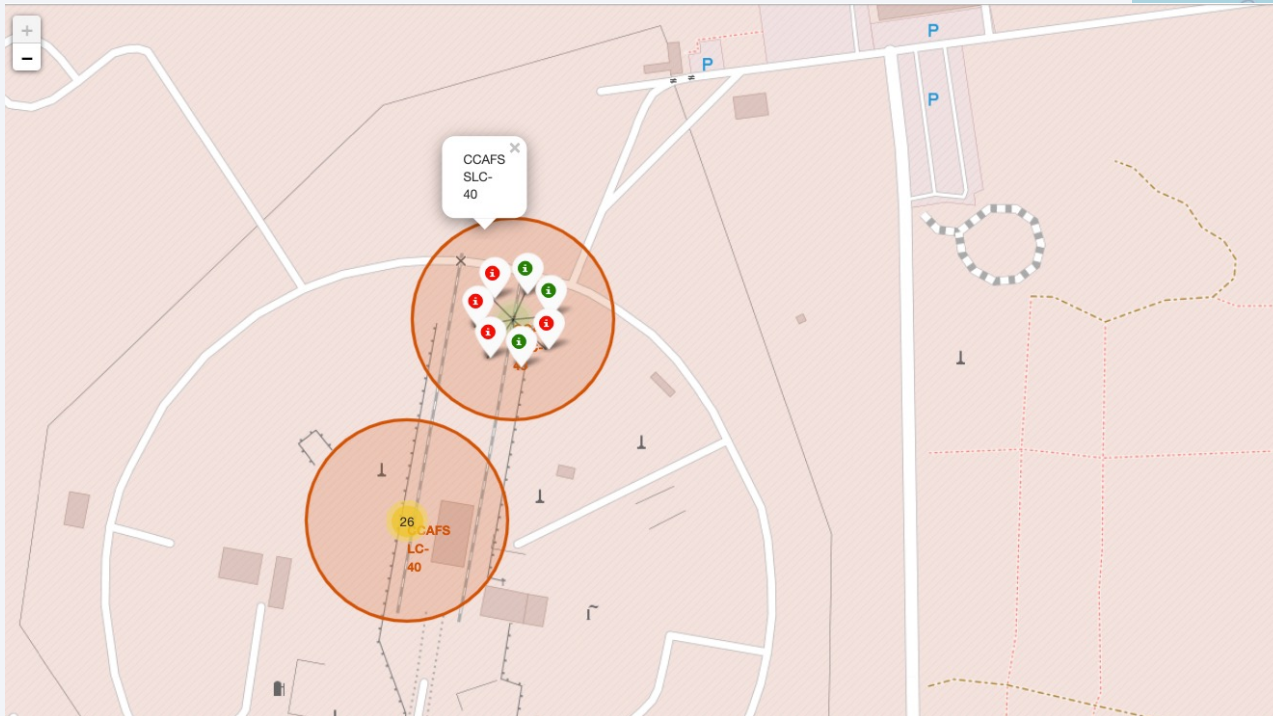
- The map on my lab never actually generated it said to goto File->Trust Notebook but that wasn't an option that existed, using the screenshot from the lab itself

* Launch Sites are situated on both US coasts near the coastline



SpaceX Launch Outcomes

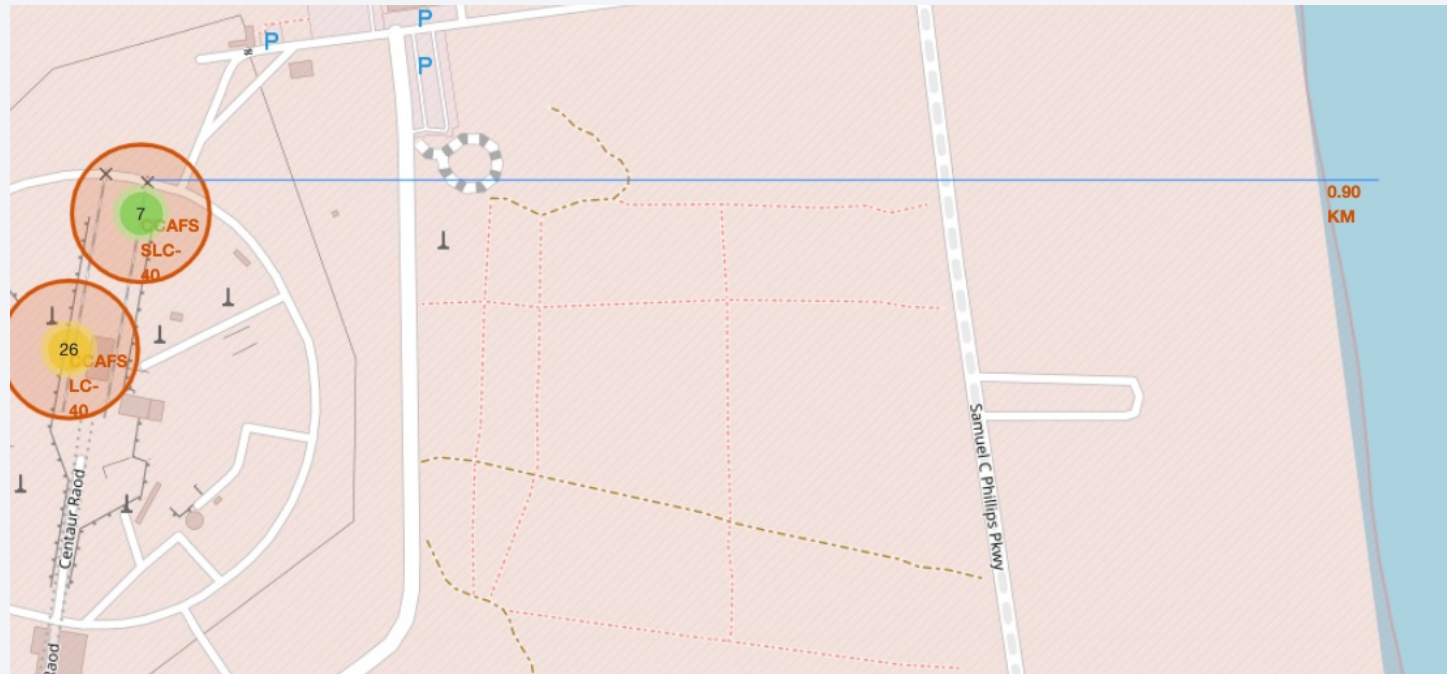
- Again my map in the lab would not display so I have to use the provided screenshots

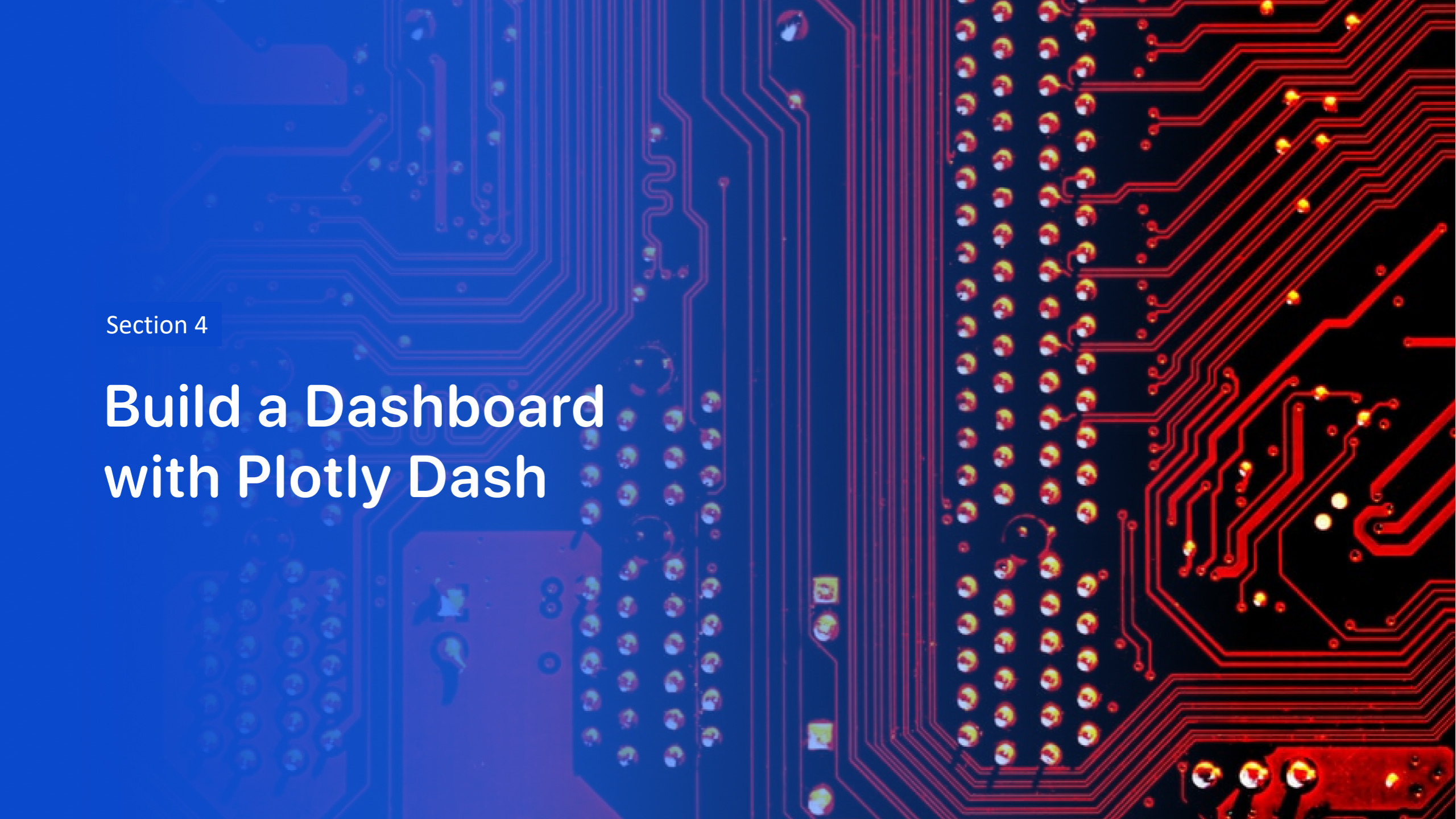


- File-> Trust Notebook does not exist!

SpaceX Launch Site Proximity to Civil Infrastructure

- Once again, the map in my lab did not display. It says to goto Files->Trust Notebook but that option does not exist. I am using a provided screenshot from the lab:



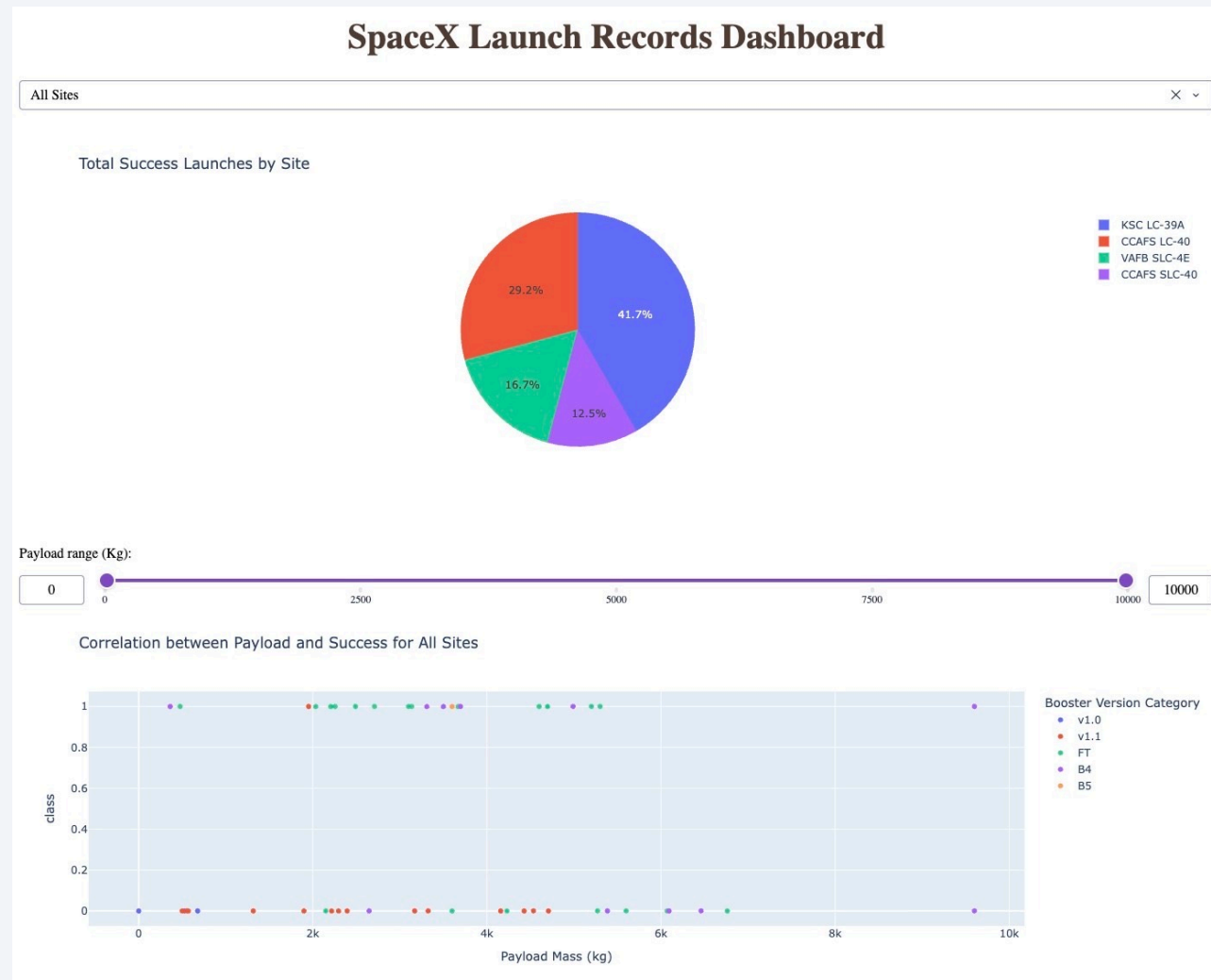


Section 4

Build a Dashboard with Plotly Dash

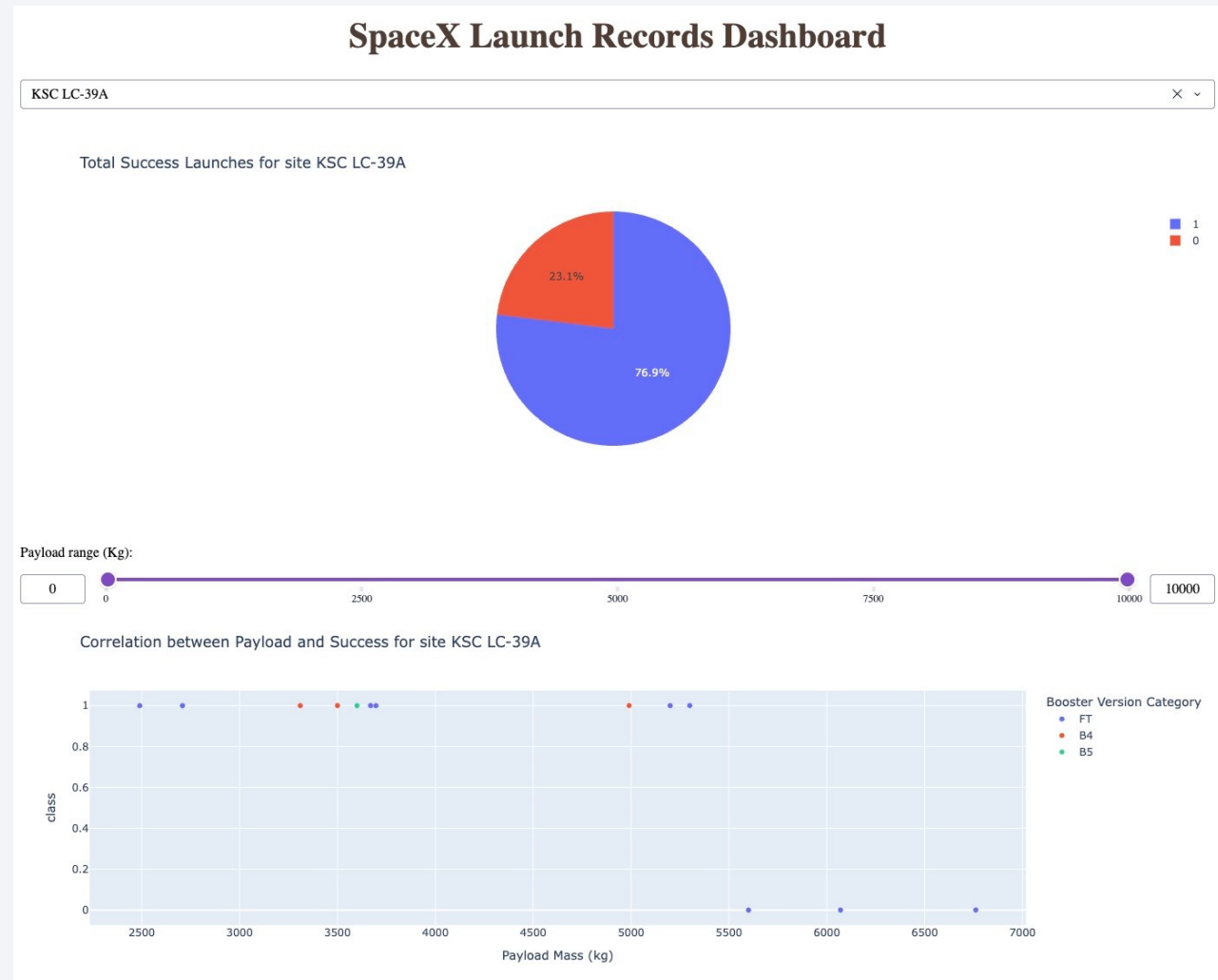
SpaceX Launch Dashboard Screenshot

- Pie Chart represents percentage of success by Launch Site
- Scatterplot shows relationship between payload mass and success rate by launch site

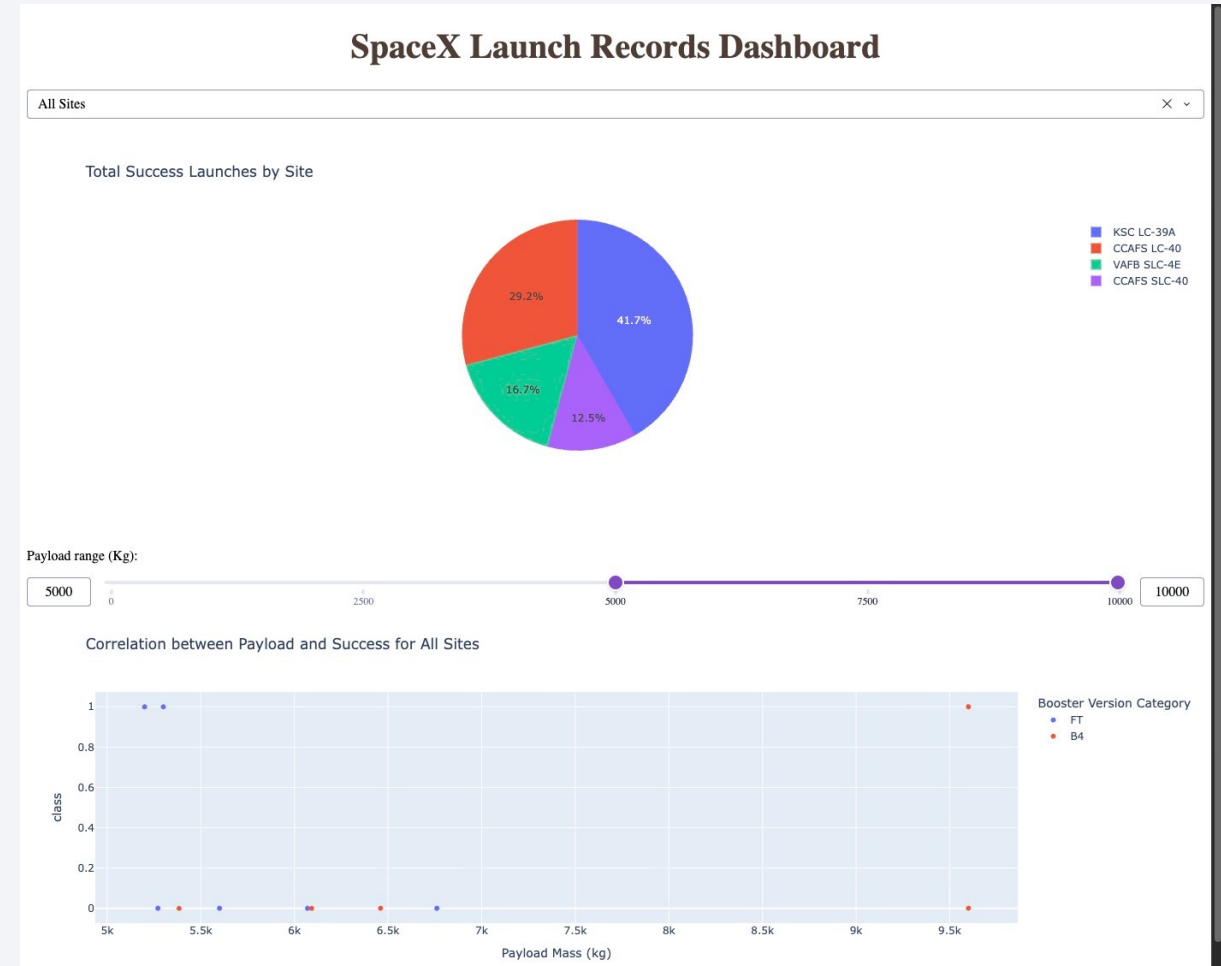
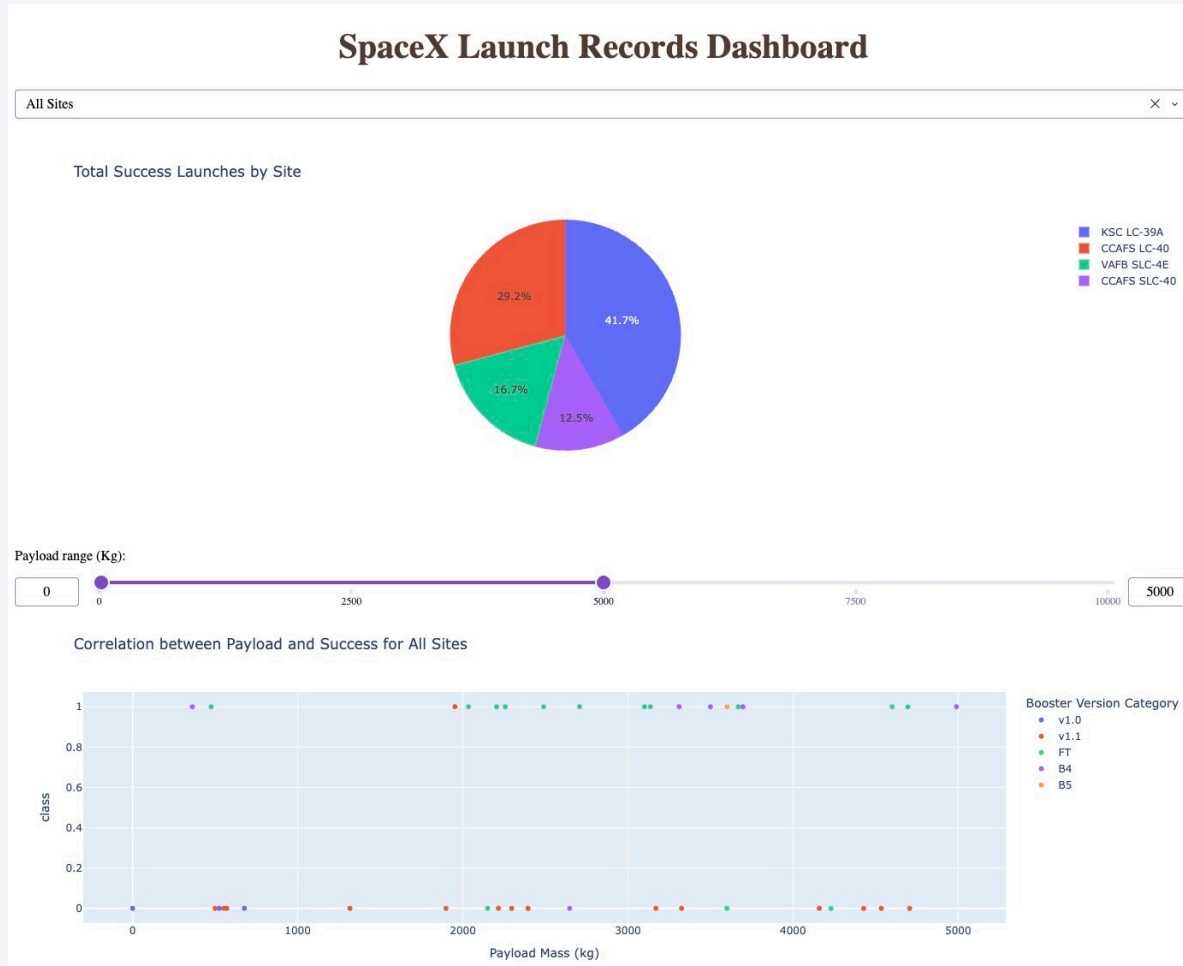


SpaceX Launch Site With Highest Success

- KSC LC-39A was the launch site with the highest success rate



Payload vs Launch Outcome Screenshot

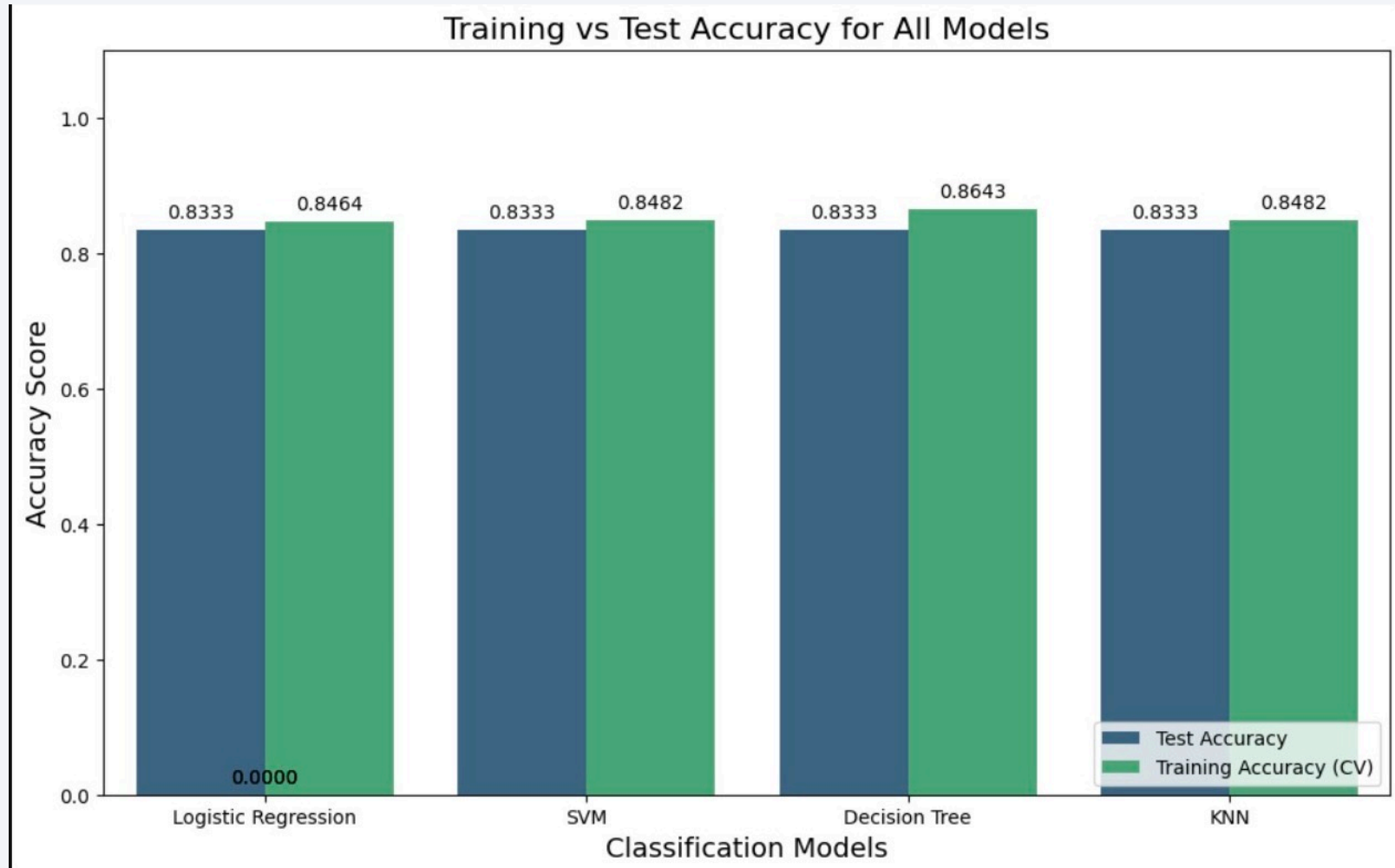


Section 5

Predictive Analysis (Classification)

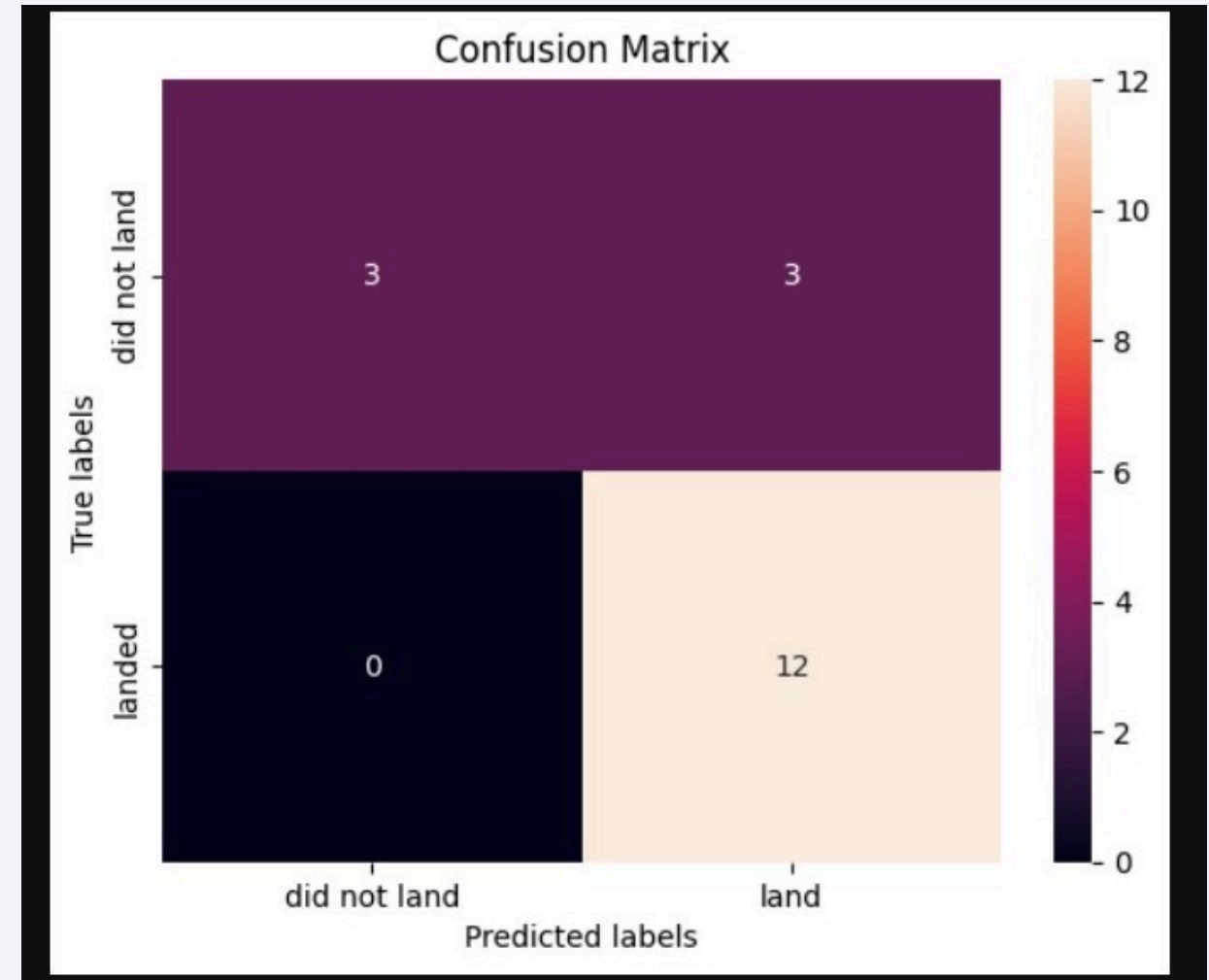
Classification Accuracy

- Decision Tree initially shows promise due to highest training accuracy when all other test accuracies were the same



Confusion Matrix

- All models performed equally well in terms of accuracy score at 83.33%
 - Via the `score(X_test, Y_test)` method
- Decision Tree performed best in training at 86.43% accuracy
- Logistic Regression, KNN, and SVM have the “best” confusion matrices (pictured on right)



Conclusions

- Experience/Practice is More Relevant Than Weight or Orbit Type
- Orbit Type is highly determinative of landing difficulty / success rate
- Machine Learning Is Highly Effective for SpaceX Launch Predictions
- KNN Is the Optimal Predictive Model for this specific application
- Folium Maps are useful to visual geographic data

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project
- All of the relevant info mentioned above is in the Jupiter lab files saved and linked to on github. See previous slides for specific URLs.

Thank you!

